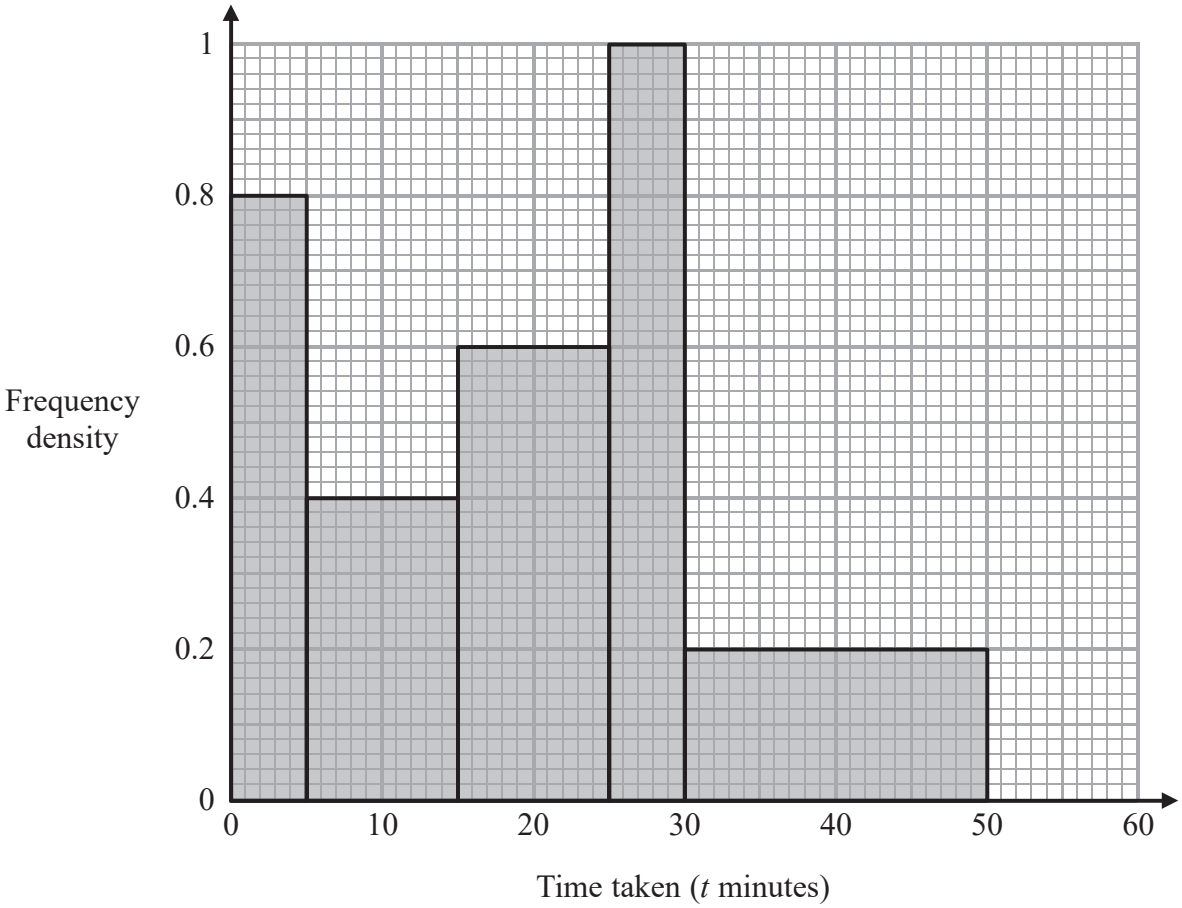
17 The histogram shows information about the times taken by some students to finish a puzzle.



(a) Complete the frequency table for this information.

Time taken (t minutes)	Frequency
$0 < t \leqslant 5$	4
$5 < t \leqslant 15$	4
$15 < t \leqslant 25$	6
$25 < t \leqslant 30$	5
$30 < t \leqslant 50$	4

(2)

Frequency = class width × frequency density.

10 × 0.4 = 4; 10 × 0.6 = 6; 5 × 1 = 5; 20 × 0.2 = 4.

1 mark
Use two different class widths

1 mark
All four frequencies correct

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(b) Find an estimate for the lower quartile of the times taken to finish the puzzle.

Total = 4 + 4 + 6 + 5 + 4 = 23 students.

Lower-quartile position = 23 ÷ 4 = 5.75.

1 mark
Find one quarter of the total

Four finished by 5 minutes. We need 1.75 of the next four.

	Count	min	
	4	10	
÷4			÷4
	1	2.5	
×1.75			×1.75
	1.75	4.375	

Assume an even spread

through the 5–15 minute class.

$Q_1 \approx 5 + 4.375$

9.375 minutes

(2)

1 mark
Correct interpolation estimate

(Total for Question 17 is 4 marks)

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